

BiocTree Graph Signal Analysis

Analysis of signal_patch_15_pct signal

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Data source: icosphere_patch_demo

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Chapter 1. Signal Analysis Results

This report presents a comprehensive analysis of graph signal data. The analysis combines spatial signal distribution visualization with Graph Fourier Transform spectral analysis.

Data Summary: • Data File: icosphere_patch_demo • Number of Vertices: 642 • Number of Edges: 1920 • Signal Analyzed: signal_patch_15_pct • Analysis Date: 2025-10-22

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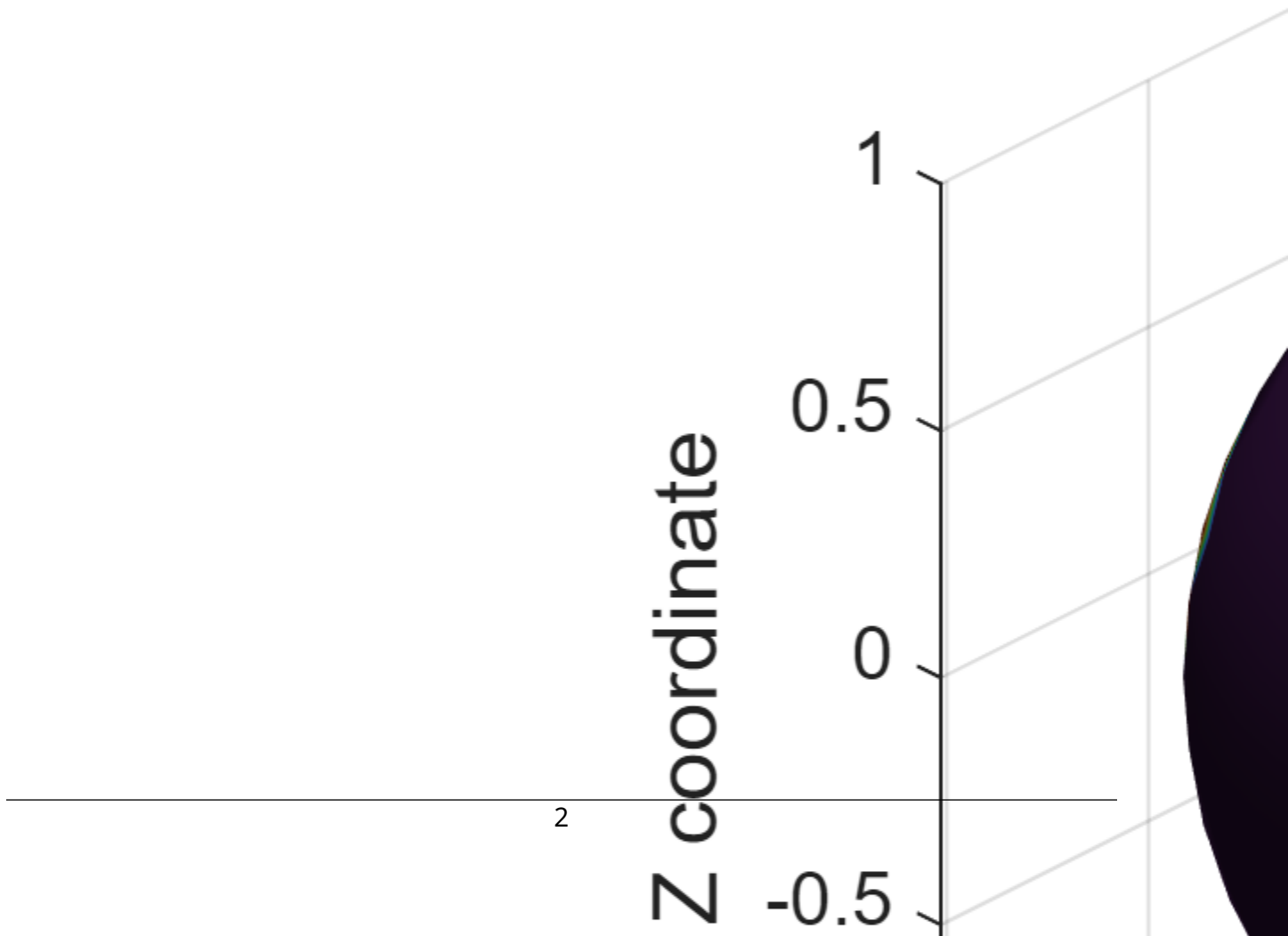


Figure: Comprehensive analysis showing spatial signal distribution (left) and spectral energy from Graph Fourier Transform (right). Low-frequency components are highlighted in red in the spectrum plot.

The surface signal visualization reveals spatial patterns in the data, while the GFT spectrum shows how signal energy is distributed across graph frequencies. Low frequencies capture smooth, global variations, while high frequencies represent localized signal features.